
a
RBP-ARK (eCLIP)
Gene search
PICALM · ENCORE/ENCORE2
QKI windows
Coordinate View: where
Experiment detail
QKI ×3 (K562 ×2, HepG2)
QC · UCSC track hub
reproducible windows .tsv
Download
BDBag / API file list
Globus (bulk)
encodeproject.org (shRNA knockdown)
shRNA RNA-seq
target = QKI
K562 / HepG2 vs control
Differential splicing
rMATS skipped exons (SE)
hg19 → lift to GRCh38
ΔPSI per exon
knockdown vs
non-targeting control
Integrate
isoforms + binding (b, c)
splicing change (d)
genome-wide rule (e)
b
150-nt exon: QKI binds upstream
RBP-ARK reproducible windows (Skipper 2.0.0), all three QKI eCLIPs · outlined = shown by gene search (log2 enrichment ≥ 3)
GENCODE v40 isoforms (− strand)
150-nt exon
(ENSE00000989276.1)
includes the exon
4 coding (MANE PICALM-202)
+2 use a 129-nt form (alt. 3′ss)
skips the exon
4 coding (e.g. PICALM-209)
← exon 12: 6.4 kb
exon 14: 6.3 kb →
eCLIP windows
QKI · K562 · ENCORE2 5007
QKI · K562 · ENCODE3
QKI · HepG2 · ENCODE3
4.1
3.3
2.4
85,990,750
85,990,500
85,990,250
85,990,000
window log2 enrichment
0
2
4
6
chr11 (GRCh38), drawn 5′ → 3′ on the minus strand
c
24-nt exon: QKI binds downstream
24-nt exon
(ENSE00001416869.1)
4 coding (e.g. PICALM-219)
8 coding (MANE PICALM-202)
← upstream exon: 3.0 kb
1.4 kb →
4.6
3.4
4.3
85,978,400
85,978,200
85,978,000
85,977,800
chr11 (GRCh38), 5′ → 3′, 11.6 kb downstream of b
d
QKI knockdown changes both exons, in opposite directions
150-nt exon (b)
24-nt exon (c)
ΔPSI +0.22
FDR <1e-10
ΔPSI +0.47
FDR <1e-10
ΔPSI -0.10
FDR <1e-10
ΔPSI -0.55
FDR <1e-10
exon inclusion (PSI)
1.0
0.5
0.0
ctl
KD
K562
log2FC -0.85
HepG2
log2FC -3.36
QKI shRNA vs control shRNA; rMATS ΔPSI and FDR, 2 replicates each (dots); log2FC = QKI mRNA (DESeq2)
Junction reads in HepG2 (2 replicates summed; not to scale)
exon 12
150-nt exon
exon 14
reads
control shRNA
incl. 101
skip 200
QKI shRNA
incl. 514
skip 123
exon 17
24-nt exon
exon 19
control shRNA
incl. 265
skip 85
QKI shRNA
incl. 82
skip 323
e
rbp-maps splice maps and the PICALM exons' own traces
upstream exon end
cassette 3′ss
cassette 5′ss
downstream exon 3′ss
QKI ENCORE2 (K562) eCLIP × HepG2 KD IP−input (×10⁻⁴)
PICALM 150-nt exon (b, d) →
peak −33 nt (max 75)
peak +60 nt (max 54)
← PICALM 24-nt exon (c, d)
−200
0
200
QKI ENCODE3 (HepG2) eCLIP × HepG2 KD IP−input (×10⁻⁴)
peak −11 nt (max 55)
peak +68 nt (max 51)
more included after KD
less included after KD
background 0.5–99.5%
50 nt exon + 300 nt intron around each splice site.
Each PICALM trace (signal at that one exon, same units) is colored by its exon's change after that knockdown. eCLIP replicate 1 shown; replicate 2 maps are saved alongside replicate 1.